Dynamics of a 3-Neuron Hopfield Network

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***Abstract*—We analyze a 3-neuron continuous-time Hopfield network using tools from nonlinear dynamical systems. The system has two stable equilibria (stored memories) and one saddle point. We establish well-posedness via global Lipschitz continuity, certify local exponential stability through linearization and quadratic Lyapunov functions, and prove input-to-state stability under bounded disturbances. Barrier functions certify disjoint forward-invariant sets around each memory. Bifurcation analysis over coupling strength reveals two saddle-node bifurcations governing memory capacity.**

I. Introduction

The Hopfield network is a recurrent neural network with symmetric coupling that models associative memory. Introduced by Hopfield [1] as one of the first rigorous applications of Lyapunov-like energy arguments to neural computation, the architecture stores patterns as stable equilibria of a continuous dynamical system.

We study the continuous-time Hopfield network:

$$\tau \dot{x} = -x + W\sigma(x) + b \tag{1}$$

where $x, b \in \mathbb{R}^n$, $W \in \mathbb{R}^{n\times n}$ is symmetric with zero diagonal, and $\sigma = \tanh$ acts element-wise. The state x represents neuron membrane potentials evolving under mutual excitation/inhibition (through $W\sigma(x)$), a passive leak term ($-x$), and a tonic bias current (b). We fix $n = 3$, which generates rich dynamical behavior — multiple attractors, bifurcations, and nontrivial basin geometry — while retaining analytic tractability. Throughout this report, we set

$$\tau = 1, \quad W = \begin{pmatrix} 0 & 2 & -1 \\ 2 & 0 & 1.5 \\ -1 & 1.5 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0.1 \\ -0.2 \\ 0.15 \end{pmatrix}. \tag{2}$$

This report summarizes the analyses performed over the course of the semester, applying the tools of nonlinear dynamics to characterize the Hopfield network's dynamics: existence and uniqueness of solutions, sensitivity to perturbations, equilibrium structure and linearized stability, Lyapunov stability and region of attraction estimates, input-to-state stability under disturbances, barrier functions for forward invariance, and bifurcation structure under parameter variation.

II. Well-Posedness

We first establish that (1) has unique solutions on all of $\mathbb{R}^n$ by verifying that the right-hand side is globally Lipschitz.

A. Global Lipschitz Constant

Define $f(x) = \tau^{-1}(-x + W\sigma(x) + b)$. The Jacobian is

$$Df(x) = \frac{1}{\tau}(-I + W\,\mathrm{diag}(\sigma'(x))). \tag{3}$$

Since $\sigma = \tanh$, we have $|\sigma'(x_i)| = |1 - \tanh^2(x_i)| \le 1$ for all x_i, with equality at $x_i = 0$. Therefore

$$\begin{aligned} \sup_x \|Df(x)\| &= \frac{1}{\tau}\|-I + W\,\mathrm{diag}(\sigma'(x))\|_2 \\ &\le \frac{1}{\tau}(1 + \|W\|_2). \end{aligned} \tag{4}$$

The operator norm is $\|W\|_2 = 3.037$ (the largest singular value of W). Since this bound is finite and independent of x, the system is globally Lipschitz with constant

$$L \le \frac{1}{\tau}(1 + \|W\|_2) = 4.037. \tag{5}$$

Numerical optimization via `scipy.optimize` confirms $L \approx 4.037$, achieved at $x = 0$ where all $\sigma'(x_i) = 1$.

By the Picard-Lindelof theorem, the system admits a unique solution $x(t)$ on $[0, \infty)$ for any initial condition $x_0 \in \mathbb{R}^3$, and by Gronwall's inequality, nearby solutions diverge at most exponentially: $\|x(t) - z(t)\| \le \|x(0) - z(0)\| e^{Lt}$.

III. Picard Iteration

To numerically validate the existence and uniqueness theorem, we compare the Picard iterates

$$x^{(k+1)}(t) = x_0 + \int_0^t f\big(x^{(k)}(s)\big)ds, \quad x^{(0)}(t) = x_0, \tag{6}$$

against the RK45 numerical solution on $t \in [0, 1]$ with $x_0 = (0.5, 0.3, 0.8)^\top$.

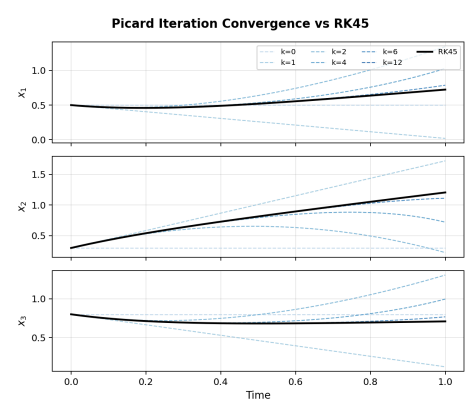

Fig. 1. Picard iteration convergence. Dashed lines show iterates $k = 0, 1, 2, 4, 6, 12$; solid black is the RK45 reference. By $k = 12$ the Picard iterate is visually indistinguishable from the true solution.

Fig. 1 shows that by 12 iterations, the Picard iterate has converged to the true solution across all three components. The convergence rate is geometric, consistent with the contraction mapping proof of Picard's theorem.

IV. Sensitivity Analysis

We quantify the sensitivity of solutions to perturbations in both initial conditions and parameters, comparing the true trajectory deviation against the analytical bounds.

A. Initial Condition Perturbation

We perturb the initial condition by $\delta x_0 = x_0/10$ and compare $\|x(t) - z(t)\|$ against the Gronwall bound $\|\delta x_0\| e^{Lt}$ on $t \in [0, 0.2]$.

B. Parameter Perturbation

We perturb the bias by $\delta b = b/10$. The perturbed system is $\dot{z} = f(z) + g(z)$ where $g(z) = \delta b/\tau$. This is a constant perturbation with $\|g(z)\| \leq \mu = \|\delta b\|/\tau \approx 0.027$. The bound from eq. 3.15 gives $\|x(t) - z(t)\| \leq (\mu/L)(e^{Lt} - 1)$.

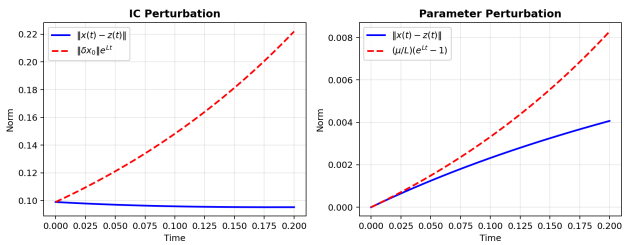

Fig. 2. Sensitivity analysis on $t \in [0, 0.2]$. Left: IC perturbation. Right: parameter perturbation. Blue: true difference. Red dashed: Gronwall bound. The bounds are conservative but correctly capture the exponential growth rate.

In both cases (Fig. 2), the analytical bounds correctly upper-bound the true deviation. The conservatism is moderate on this short time interval; on longer intervals the exponential bound becomes increasingly loose. Moreover, since we start in the region of attraction near a stable equilibrium, the IC-perturbed trajectory converges to that equilibrium, and therefore has decreasing distance to the unperturbed trajectory. This motivates the Lyapunov-based local analysis in subsequent sections.

V. Equilibrium Analysis

A. Identifying Equilibria

Equilibria satisfy $f(x^*) = 0$, i.e., $x^* = W \tanh(x^*) + b$. We perform multi-start root-finding: 80 random initial guesses in $[-4, 4]^3$ are passed to `scipy.optimize.fsolve`, keeping only solutions with $\|f(x^*)\| < 10^{-10}$, and deduplicating by rounding to 4 decimal places. This yields three equilibria:

Equilibrium	**Value**	$\lambda(Df(x^*))$	**Type**
x_0^*	$(-1.47, -2.61, -0.43)$	$-1.51, -0.90, -0.59$	Stable
x_1^*	$(-0.15, 0.06, 0.39)$	$-3.90, 1.06, -0.15$	Saddle
x_2^*	$(1.45, 2.53, 0.74)$	$-1.46, -0.65, -0.89$	Stable

The two stable nodes (x_0^* and x_2^*) serve as stored memories of the Hopfield network. The saddle point x_1^* lies on the basin boundary separating their domains of attraction.

B. Linearization at x_0^*

At x_0^*, the Jacobian is

$$\begin{aligned} A &= Df(x_0^*) \\ &= \frac{1}{\tau}(-I + W \operatorname{diag}(\sigma'(x_0^*))) \\ &= \begin{pmatrix} -1.000 & 0.382 & -0.191 \\ 0.043 & -1.000 & 0.032 \\ -0.833 & 1.249 & -1.000 \end{pmatrix}. \end{aligned} \tag{7}$$

All eigenvalues ($\lambda_1 = -1.505$, $\lambda_2 = -0.901$, $\lambda_3 = -0.594$) are real and strictly negative. Since all eigenvalues have nonzero real part, the Hartman–Grobman theorem applies: the nonlinear flow near x_0^* is topologically conjugate to the linearized flow $\delta\dot{x} = A\delta x$. By Theorem 8.3 [2], x_0^* is exponentially stable.

C. Linearization at a Non-Equilibrium

At the origin $x_0 = (0, 0, 0)$, which is *not* an equilibrium since $f(0) = b \neq 0$, the first-order expansion gives $\delta\dot{x} \approx A_0 \delta x + C$ where $A_0 = -I + W$ and $C = f(0) = b$. Hartman–Grobman does not apply here since the origin is not an equilibrium point.

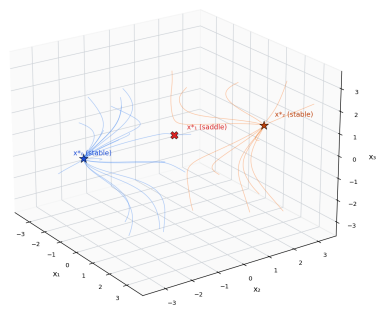

Fig. 3. 3D phase portrait showing trajectories from random initial conditions converging to the two stable equilibria x_0^* (blue) and x_2^* (orange). The saddle x_1^* (red X) sits on the basin boundary.

VI. Lyapunov Stability and Region of Attraction

A. Lyapunov Function Construction

We construct a quadratic Lyapunov function for x_0^* by solving the continuous-time Lyapunov equation (CTLE):

$$A^\top P + PA = -Q = -I_3 \tag{8}$$

where $A = Df(x_0^*)$. Since A defines an globally exponentially stable system, (8) has a unique positive definite solution:

$$P = \begin{pmatrix} 0.575 & 0.078 & -0.236 \\ 0.078 & 0.513 & 0.300 \\ -0.236 & 0.300 & 1.071 \end{pmatrix}, \tag{9}$$

with eigenvalues $\lambda_{\min(P)} = 0.277$ and $\lambda_{\max(P)} = 1.254$. The candidate Lyapunov function is

$$V(x) = \delta x^\top P \delta x, \quad \delta x = x - x_0^*. \tag{10}$$

Near x_0^*, the linearized dynamics give $\dot{V} \approx \delta x^\top (A^\top P + PA)\delta x = -\|\delta x\|^2 < 0$, confirming local exponential stability per Theorem 11.1.

B. Region of Attraction Estimate

The region of attraction (RoA) is the largest set from which trajectories converge to x_0^*. We approximate it by finding the largest Lyapunov sublevel set $\Omega_c = \{x : V(x) \le c\}$ on which $\dot{V} < 0$.

We sample 5000 directions uniformly on S^2 and, for each direction, perform a binary search for the distance from x_0^* at which $\dot{V}$ transitions from negative to non-negative. The minimum across all directions yields a radius of $r = 3.06$, corresponding to a sublevel value of $c = 6.13$.

By Proposition 7.1, Ω_c is forward invariant when $\dot{V} \le 0$ on Ω_c. By LaSalle's invariance principle (Theorem 10.2), since the only invariant set within $\left\{x \in \Omega_c : \dot{V}(x) = 0\right\}$ is $\{x_0^*\}$ itself (verifiable by examining the structure of the Jacobian), trajectories converge to x_0^*.

VII. Input-to-State Stability

We introduce an additive disturbance $d(t) \in \mathbb{R}^3$ into the Hopfield network:

$$\dot{x} = \frac{1}{\tau}(-x + W\sigma(x) + b) + d(t), \quad \|d\|_\infty < \infty. \tag{11}$$

Here $\|d\|_\infty = \text{ess sup}_{t \ge 0} \|d(t)\|$. Neurally, $d(t)$ represents unmodeled current injection — synaptic input from external neurons, sensory drive, or channel noise. We analyze ISS using the Lyapunov function $V(x) = \delta x^\top P \delta x$ from Section VI.

A. ISS Certificate

The Lyapunov derivative along the disturbed flow is:

$$\dot{V} = 2\delta x^\top P[f(x) + d(t)]. \tag{12}$$

Near x_0^*, the undisturbed component satisfies $2\delta x^\top P f(x) \approx -\delta x^\top Q \delta x = -\|\delta x\|^2$. The cross-term is bounded by Cauchy–Schwarz:

$$|2\delta x^\top P d| \le 2\|P\|\|\delta x\|\|d\|, \tag{13}$$

where $\|P\| = \lambda_{\max(P)} = 1.254$. Combining:

$$\dot{V} \le -\|\delta x\|^2 + 2\|P\|\|\delta x\|\|d\|. \tag{14}$$

To obtain an ISS Lyapunov function per Theorem 11.2, we need $\dot{V} \le -\alpha_3(\|\delta x\|)$ when $\|\delta x\| \ge \rho(\|d\|)$, not merely $\dot{V} < 0$. We split the decay term:

$$\dot{V} \le -\frac{1}{2}\|\delta x\|^2 - \frac{1}{2}\|\delta x\|^2 + 2\|P\|\|\delta x\|\|d\|. \tag{15}$$

Define $\rho(s) = 4\|P\|s$. When $\|\delta x\| \ge \rho(\|d\|)$:

$$\begin{aligned} 2\|P\|\|d\| &\le \frac{1}{2}\|\delta x\| \\ \implies -\frac{1}{2}\|\delta x\|^2 + 2\|P\|\|\delta x\|\|d\| &\le 0 \end{aligned} \tag{16}$$

Therefore:

$$\|\delta x\| \ge \rho(\|d\|) \implies \dot{V} \le -\frac{1}{2}\|\delta x\|^2, \tag{17}$$

establishing that V is an ISS Lyapunov function with $\alpha_3(r) = r^2/2$ and $\rho(s) = 4\|P\|s \approx 5.02s$. By Corollary 11.1, the system is E-ISS.

B. Numerical Verification

We simulate (11) with sinusoidal disturbances $d(t) = a[\sin 2t, \cos 3t, \sin t]^\top$ for $a \in \{0, 0.05, 0.10, 0.15, 0.30\}$. For this signal, $\|d\|_\infty = a \cdot \sup_t \|[\sin 2t, \cos 3t, \sin t]\|_2 = 1.586a$ (computed numerically).

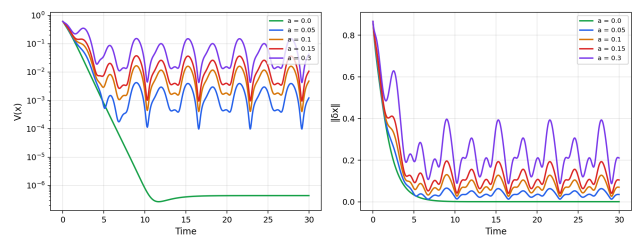

Fig. 4. ISS trajectories. Left: $V(t)$ on log scale. Right: $\|\delta x\|$. Undisturbed trajectory ($a = 0$, green) converges to x_0^*. Disturbed trajectories oscillate in neighborhoods whose size scales with $\|d\|_\infty$.

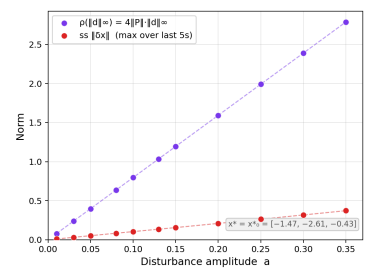

Fig. 5. ISS gain scaling. Purple: analytical bound $\rho = 4\|P\| \cdot \|d\|_\infty$. Red: actual steady-state $\|\delta x\|$ (max over last 5 seconds of simulation). Both scale linearly with a, confirming the linear class $\mathcal{K}$ gain. The analytical bound is approximately $11\times$ conservative.

Fig. 4 and Fig. 5 confirm the ISS property. The steady-state error scales linearly as $\approx 0.45\|d\|_\infty$, much tighter than the analytical bound $\rho(\|d\|_\infty) = 5.02\|d\|_\infty$. The conservatism arises from two sources, aside from the theory providing an upper bound: the linearization approximation $2\delta x^\top P f(x) \approx -\|\delta x\|^2$ and the half-splitting step used to extract α_3.

VIII. Barrier Functions and Forward Invariance

We construct barrier functions to certify forward invariance of safe sets around the stable equilibria, following the framework of Lecture 12.

A. Ball Barrier

Define $h(x) = r^2 - \|x - x^*\|^2$, so the safe set is $\mathcal{S} = \{x : h(x) \ge 0\}$, a ball of radius r centered at x^*. This is the 0-superlevel set of h in the sense of Definition 12.2. The barrier function condition (Definition 12.6) for forward invariance requires:

$$\dot{h}(x) \ge -\alpha(h(x)), \quad \forall x \in \mathcal{S}, \quad \alpha \in \mathcal{K}_\infty^e. \tag{18}$$

On the boundary $\partial\mathcal{S}$ where $h = 0$, this reduces to the Nagumo condition $\dot{h} \geq 0$ (Theorem 12.2, Case II). Computing:

$$\dot{h} = -2(x - x^*)^\top f(x). \tag{19}$$

We verify this numerically by sampling 10,000 unit directions $\hat{u}$ uniformly on S^2. For each direction, binary search finds the largest r such that $\dot{h}(x^* + r\hat{u}) \geq 0$. The minimum across all directions gives the certified radius — the largest ball that is provably forward invariant.

Results: $r = 2.947$ at x_0^* and $r = 2.865$ at x_2^*. The two balls are disjoint: $\|x_0^* - x_2^*\| = 6.03 > 2.947 + 2.865 = 5.81$, with the saddle x_1^* sitting in the gap between them.

B. Lyapunov Sublevel Barrier

By Proposition 12.1, the function $h(x) = c - V(x)$ defines the Lyapunov sublevel set $\Omega_c = \{x : V(x) \leq c\}$ as a barrier set. Since $\dot{V} < 0$ inside Ω_c, we have $\dot{h} = -\dot{V} > 0$, which is strictly stronger than the barrier condition $\dot{h} \geq -\alpha(h)$. The maximum sublevel where $\dot{V}$ remains strictly negative is $c = 6.13$.

Both methods certify valid forward-invariant sets around x_0^*, but they characterize different regions – neither strictly contains the other (Fig. 7). The ball barrier is a sphere of radius 2.95; Ω_c is an ellipsoid whose semi-axes (2.21 to 4.70) are aligned with the eigenvectors of P. In the direction of P's largest eigenvector (where the ellipsoid is narrowest), the ball extends somewhat further. The ellipsoid is larger in total volume by roughly 35% and covers more of state space in most directions.

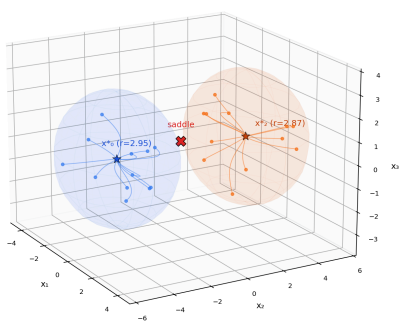

Fig. 6. Ball barriers around both stable equilibria. Blue: x_0^* ($r = 2.95$). Orange: x_2^* ($r = 2.87$). Trajectories starting inside each ball remain inside and converge. The saddle (red X) sits in the gap between the two disjoint balls.

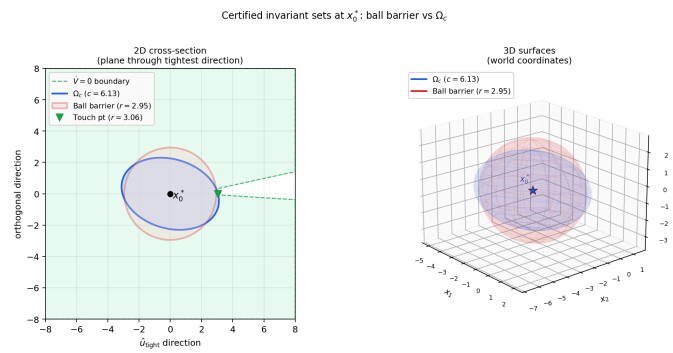

Fig. 7. Comparison of certified invariant sets at x_0^*. *Left:* 2D cross-section in the plane containing the tightest direction $\hat{u}$; the green shading shows $\{\dot{V} < 0\}$, which extends well beyond the plot in most directions. The touch point (triangle) marks where Ω_c just meets the $\dot{V} = 0$ boundary. *Right:* 3D view of the ball barrier (red, $r = 2.95$) and Lyapunov ellipsoid Ω_c (blue, $c = 6.13$) in world coordinates.

IX. Bifurcation Analysis

We parameterize the weight matrix as $W \to \alpha W$ and sweep $\alpha \in [0, 2]$ to study how the equilibrium structure changes with coupling strength.

A. Bifurcation Structure

For each α in a grid of 300 values, we repeat the multi-start root-finding procedure from Section V (80 random initial guesses, fsolve, deduplication, eigenvalue-based stability classification).

For small α, the leak term $-x$ dominates and $g(x) = \alpha W \tanh(x) + b$ is a contraction. As α increases, the feedback through W becomes strong enough to sustain self-reinforcing saturated states:

- $\boldsymbol{\alpha \approx 0.55}$**:** A saddle-node bifurcation creates two new equilibria (one stable, one saddle).The network can now store one extra memory.
- $\boldsymbol{\alpha \approx 1.20}$**:** A second saddle-node bifurcation creates two more equilibria, bringing the total to 5 (2 stable, 3 unstable).

At the nominal parameter $\alpha = 1$, the system has 3 equilibria (2 stable, 1 unstable), consistent with all analyses in previous sections.

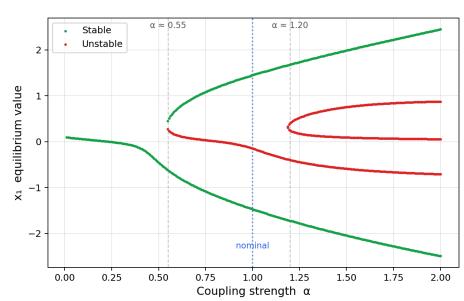

Fig. 8. Bifurcation diagram. Green: stable equilibria. Red: unstable equilibria. Dashed lines mark the two saddle-node bifurcations. Blue dotted line: nominal $\alpha = 1$.

B. Interpretation

The memory capacity of the network is directly controlled by the coupling strength α. The classical Hopfield result that an n-neuron network can store approximately $0.14n$ patterns corresponds to the regime where coupling is strong enough to support multiple stable equilibria without destabilizing all of them. In our 3-neuron system with the given W structure, the maximum memory capacity is 2 (achieved for $\alpha > 1.20$). Stronger coupling creates more attractors but also more unstable equilibria (basin boundaries), and the basin geometry becomes increasingly complex.

X. Conclusion

We have demonstrated this low-dimensional 3-neuron Hopfield network exhibits rich nonlinear phenomena. The key results are:

1) **Well-posedness:** Global Lipschitz continuity ($L = 4.037$) guarantees unique solutions for all time, validated by Picard iteration convergence.
2) **Equilibrium structure:** Three equilibria: two stable nodes (stored memories) and one saddle (basin boundary), with exponential stability certified by linearization and the Hartman–Grobman theorem.
3) **Lyapunov analysis:** A quadratic Lyapunov function from the CTLE certifies local exponential stability with an RoA radius of 3.06, and LaSalle's invariance principle (Theorem 10.2) upgrades stability to asymptotic convergence.
4) **ISS:** The system is E-ISS under bounded disturbances (Corollary 11.1). The analytical bound is approximately $11\times$ conservative compared to numerical steady-state behavior.
5) **Barrier functions:** Ball barriers (Theorem 12.2) certify forward-invariant spheres of radius 2.95 and 2.87 around the two memories. Lyapunov sublevel barriers (Proposition 12.1) certify an ellipsoidal set ($c = 6.13$, semi-axes 2.21–4.70). Both are valid but neither contains the other; the ellipsoid covers 35% more volume while the ball extends further in P's largest-eigenvector direction.
6) **Bifurcation:** Two saddle-node bifurcations at $\alpha \approx 0.55$ and $\alpha \approx 1.20$ control memory capacity, connecting the network's computational function to its coupling strength.

All source code and figure generation scripts are available at https://github.com/gavin-hyl/hopfield-3d.

References

[1] J. J. Hopfield, "Neural networks and physical systems with emergent collective computational abilities," *Proceedings of the National Academy of Sciences*, vol. 79, no. 8, pp. 2554–2558, 1982.

[2] A. D. Ames, *Nonlinear Dynamics and Control*. Unpublished lecture notes, CDS 232, California Institute of Technology, 2026.